Project 4B - Wokwi NTC Thermistor Temperature Test Bench

Purpose

Validate a simulated NTC thermistor temperature measurement workflow using Arduino ADC data, sensor conversion math, threshold classification, Python analysis, plots, and automated report generation.

Circuit Description

The Wokwi simulation uses an Arduino Uno analog input connected to an NTC temperature sensor model. The Arduino reads A0, converts ADC counts to voltage, estimates thermistor resistance, converts resistance to temperature using the Beta equation, and prints CSV data to the serial monitor.

Formula Summary

ADC to voltage	voltage = adc_value * 5.0 / 1023.0
Voltage to resistance	resistance_ntc = fixed_resistor * voltage / (5.0 - voltage)
Beta equation	1/T = 1/T0 + (1/B) * ln(R/R0)
Kelvin to Celsius	temperature_c = temperature_k - 273.15

Test Limits

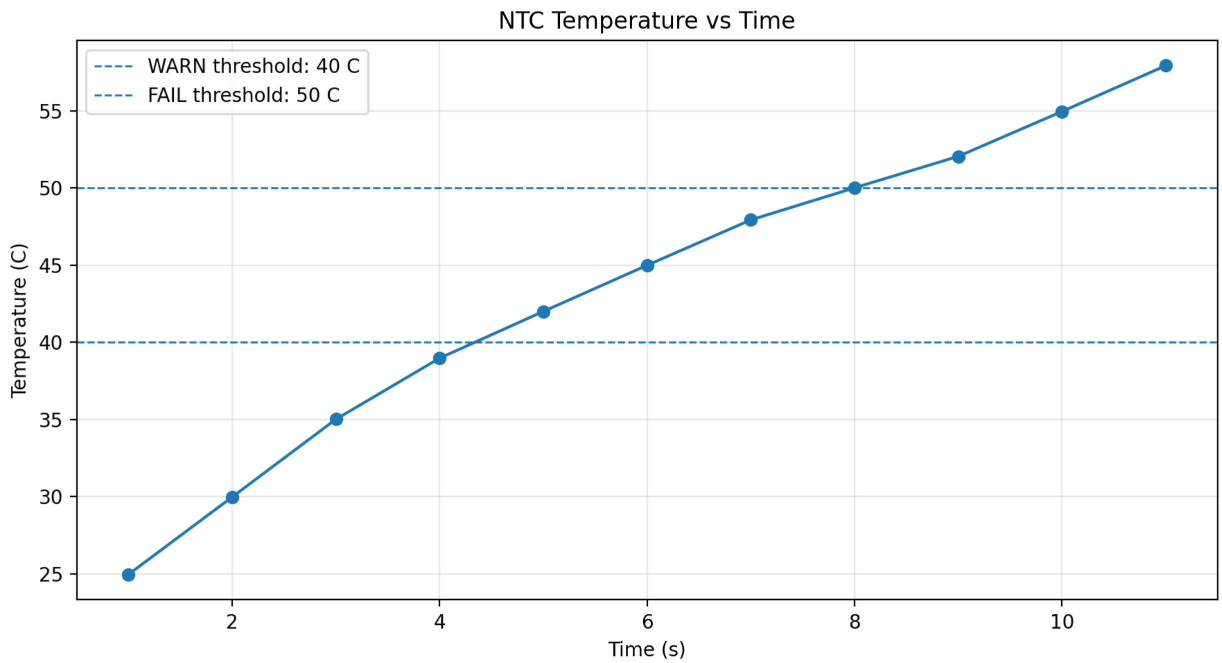
Status	Temperature Condition
NORMAL	temperature < 40 C
WARN	40 C <= temperature < 50 C
FAIL	temperature >= 50 C

Engineering Summary

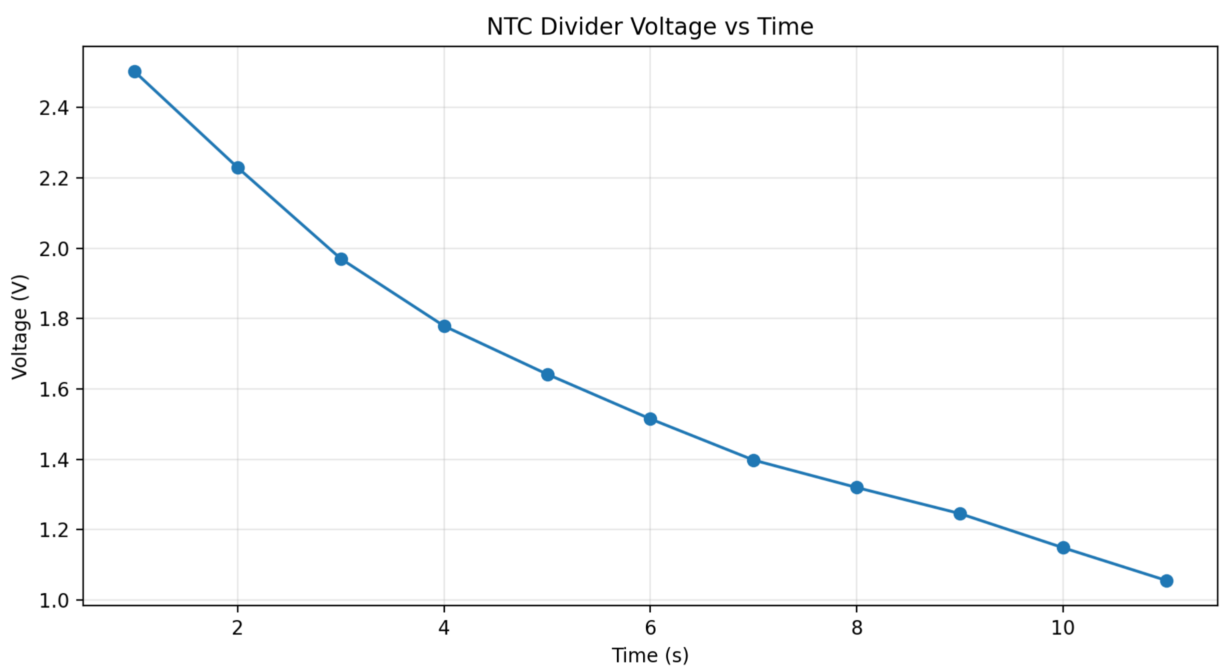
Metric	Value
Samples analyzed	11
NORMAL count	4
WARN count	3
FAIL count	4
Min temperature	24.96 C
Max temperature	57.94 C
Average temperature	43.53 C
Min voltage	1.056 V
Max voltage	2.502 V
Min resistance	2676.6 ohms
Max resistance	10019.6 ohms
Final result	FAIL

Generated Plots

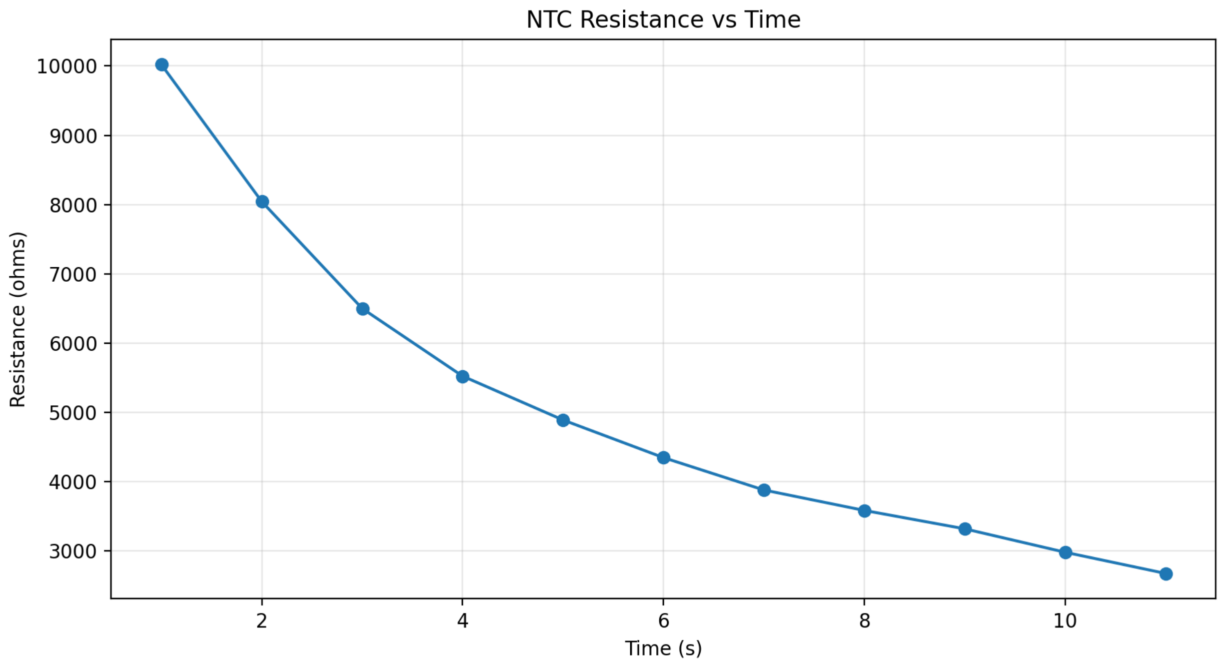
Temperature Plot



Voltage Plot



Resistance Plot



Conclusion

This project connects circuit theory, embedded ADC reading, thermistor sensor math, threshold validation, data logging, Python automation, plot generation, and PDF reporting. It is stronger portfolio evidence than a simple potentiometer ADC demo because it models a realistic sensor validation workflow.